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An Experimental Study: The Restorative Effect of Soundscape Elements in a Simulated Open-Plan Office

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Summary
Studies have proven that restorative environments show apparent positive effects on human health and wellbeing. In addition to the restorative effects of visual environments, recent research has shown increased interest in restorative soundscapes. The aim of this study is to examine whether a soundscape element perceived as pleasant has restorative effects in a simulated open-plan office. Subjects (N = 75) were recruited to participate in three experiments under different conditions: (a) different sound types, (b) different sound sequences, and (c) different audio-visual conditions. To measure stress recovery and fatigue mitigation, seven indicators representing participants' physiological responses, psychological experience and cognitive performance were used. The results showed that the soundscape elements that were perceived as pleasant had an apparent positive effect on tiredness restoration and annoyance reduction. Interestingly, sound elements had a greater effect on psychological restoration compared with visual scenes under the simulated conditions of an open-plan office. Furthermore, it was shown that continuous sound did not have better restorative effects than intermittent sound. These findings revealed the restorative potential of a good soundscape and might provide some practical guidance on how to create a restorative environment in open-plan offices.

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1. Introduction

In today's society, people suffer from increasing stress and fatigue in their daily lives and busy work, which threatens their health and wellbeing [1]. In open-plan offices, a very popular and important office style in the past decade, researchers have identified a variety of social and physical environmental problems, such as crowding and noise. The presence of these problems in open-plan offices results in more physical and psychological stress and poorer work performance as compared to traditional offices [2, 3, 4]. Therefore, restoration is very important and necessary for people working in open-plan offices.

The term "restoration", derived from environmental psychology studies, is defined as the process of renewing, recovering, or reestablishing physical, psychological and social resources or capabilities diminished in ongoing efforts to meet adaptive demands [5, 6, 7]. Based on empirical studies, exposure to restorative environments, such as natural parks, is proven to effectively provide positive effects on physical responses (e.g., heart rate and skin conductance level) [8], psychological experience (e.g., tension and annoyance) [9], and cognitive performance (e.g., sustained attention tests) [10, 11]. Based on these studies, a restorative environment may be useful to help people restore from stress and fatigue in open-plan offices.

However, most studies on the restorative environment have focused on its visual aspects [12, 13, 14]. A number of studies have proven that visual factors in natural environments (e.g., trees, lawns, and rivers) can promote better restoration than those in built environments [15, 16]. Nonetheless, considering that a person's perception is based on multiple senses, including visual and auditory senses, study of restorative environments has broadened to include not only visual perceptions, but also environment sounds [17, 18, 19]. Similar to the findings involving visual aspects, natural sounds, such as birdsong and water sounds, are found to provide better restorative effects than artificial noise [20, 21, 22]. A questionnaire survey further investigated how a specific natural sound (i.e., birdsong) was associated with restoration [23]. In addition, the relationships between soundscape's subjective evaluation and restorative measurements were preliminarily explored in a physiological investigation [24]. Moreover, Payne developed and tested a Perceived Restorative Soundscape Scale (PRSS), which was designed to assess perceptions of a soundscape's potential to provide psychological restoration [25]. Audio-visual interactions in restorative environments also have been explored by some researchers, who suggest that soundscape elements significantly contribute to acoustic comfort and overall visual

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landscape preference, and vice versa [26, 27, 28]. These studies indicate that soundscape elements have restorative potential on people's health and wellbeing. Therefore, in order to guide future soundscape design in open-plan offices, it is necessary to systematically examine their restorative effects on people's physical and psychological health and work performance.

The aim of this study is to explore the restorative effect of soundscape elements in a simulated open-plan office. Three experiments were conducted to answer the following questions: 1) Do the soundscape elements preferred by people have a restorative effect on emotional stress and attentional fatigue; 2) How can soundscape elements be added into original environments to fulfill and maintain their restorative effects; and 3) What are the effects of audio-visual interaction on perceived restoration in a simulated open-plan office?

2. Methodology

2.1. Pilot study

In the pilot survey, 38 sounds that were evaluated to be the most frequent sounds heard by people in their daily lives were selected. All the sounds were recorded or downloaded from the British Library Sound Archive and the Positive Soundscape Project database. From these recordings, sound samples, with a duration of 20 s and sound level of 55 dBA were made to be used as the sound stimuli for the following experiments. Thirty participants took part in the pilot experiment and were asked to evaluate their preference on 38 sound stimuli using a 7-point numerical scale. As a result, 38 sound stimuli were sorted according to their median values of preference as positive or negative sounds.

2.2. Participants

Three groups of participants attended three different experiments and the number of the participants was 30, 15 and 30. All 75 participants were graduate students of Tianjin University with different research background such as architecture, urban planning, chemical engineering, and material science. The average age of the participants was 25 years old. All of the participants had normal hearing, and their working status and stress level were very similar with young employees who worked in open-plan offices. In each experiment, the ratio of the male and female participants was kept as 1:1.

2.3. Stimuli

Experiment 1: According to the pilot study on preferences of different sounds, five sounds such as flowing water sound, birdsong, footsteps, traffic noise and air-conditioner noise, were selected as the sound stimuli used in the experiment. There are two reasons to use those sounds as the stimuli in this experiment: 1) People's attitude to those sounds ranged from negative to positive as shown in Table I. Flowing water sound and birdsong were evaluated to be the most pleasant sounds by people, whereas footsteps, traffic noise and air-conditioner noise were classified as unpleasant sounds. 2) Those sounds are popular or very easy to be added in the environment of an open-plan office. In addition, a preliminary test was carried out to decide what was the suitable sound level for sound stimuli broadcasted in the following experiments in order that subjects could hear the sound stimuli clearly but do not feel annoyed. The sound level of all the five sound sources (flowing water sound, birdsong, footsteps, traffic noise and air-conditioner noise) varied from 40 dBA to 60dBA with a step of 5 dBA to simulate the indoor sound level in open-plan offices. 15 participants (7 males and 8 females) aged 23-27 years old were exposed to the sound stimuli using headphones (AKG K702), and they were asked to select the most suitable sound level under which all the sound stimuli could be heard clearly but were not too noisy. As a result, 55dBA was chosen to be the most appropriate sound level. Therefore the sound level of all the sound stimuli used in the formal experiments was set at 55 dBA and the duration of each sound was set as 3 min according to the experiment tasks.

Table I. Preference score of five sounds with sound level of 55 dBA. AN, TN, FS, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively.

Preference	AN	TN	FS	BS	WS
Median value	−2.00	−1.00	−0.50	2.50	3.00

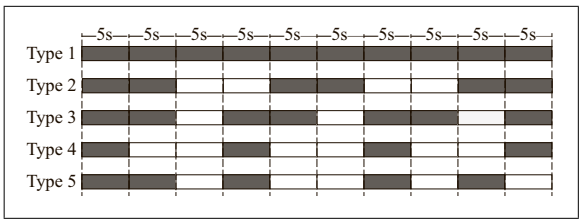

Figure 1. Configuration of sound sequences used in experiment 2, including Type 1 (continuous water sound), Type 2 (intermittent water sound, 10 s of water sound and 10 s of silence), Type 3 (intermittent water sound, 10 s of water sound and 5 s of silence), Type 4 (intermittent water sound, 5 s of water sound and 10 s of silence), and Type 5 (intermittent water sound, random duration for water sound and silence).

Experiment 2: Considering the convenience and possibility of adding sound elements into the environment of open-plan offices, flowing water sound was used as the sound stimuli in experiment 2. Figure 1 shows that the five types of sound sequences consisted of water sound and silence. Type 1 is consistent water sound, and the other types were intermittent sounds of 10-second or 5-second water sounds with a sound pressure level of 55 dBA. The interval between the intermittent water sounds was set purposely, and each sound sequence lasted for 50 seconds.

Experiment 3: In the third experiment, both air-conditioner noise and flowing water sound were used as nega-

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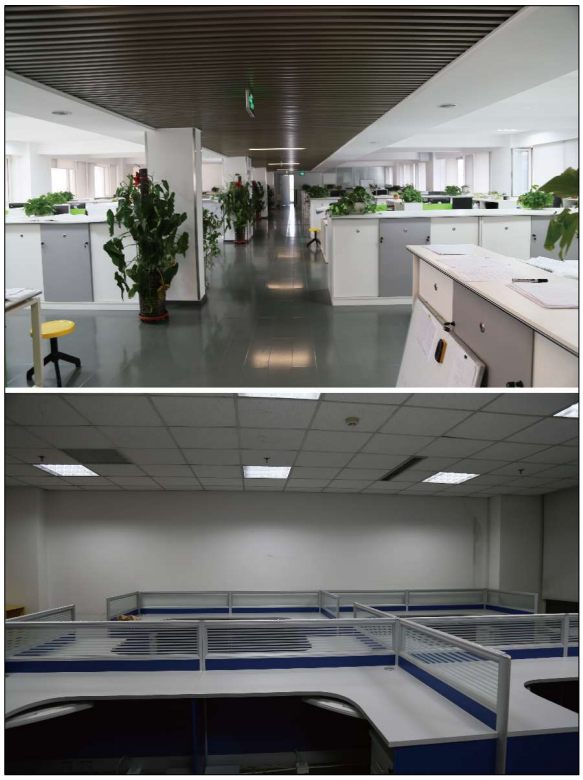

Figure 2. Two visual stimuli used in experiment 2. The upper and bottom images are a good and bad image, respectively.

tive and positive acoustic stimuli, respectively, to match different visual scenes. Two photos selected by participants through a preliminary survey as the best and the worst visual scenes from 120 photos taken in different open-plan offices were adopted as the visual stimuli, as shown in Figure 2. Altogether, 4 audio-visual stimuli were made with a time duration of 3 min, including W+GI (water sound and good image), W+BI (water sound and bad image), AN?GI (air-conditioner noise and good image) and AN+BI (air-conditioner noise and bad image).

2.4. Experiment tasks and measures

Before exposure to each experimental stimuli, participants were asked to take a 5-min calculation task which was used to create stress and attentional fatigue. The detailed calculation tasks included asking participants to perform continuous subtraction from 1895 with a step of 13. If the participants did a miscalculation, they would be asked to stop and start their calculation from the beginning. This method has been proven to be an effective way to raise people's stress and attentional fatigue in former studies [8].

As for the indicators of participants' stress and fatigue level, three aspects were considered. 1) Physiological response included systolic pressure, diastolic pressure and heart rate. Real-time data of physiological measures were obtained using a wrist blood pressure monitor. 2) Psychological measures were assessed by a 3-item self-report

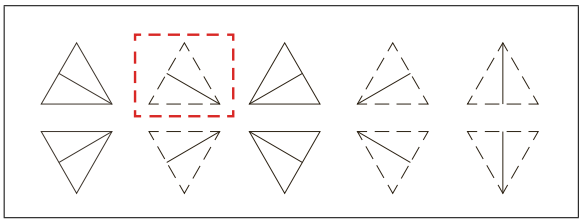

Figure 3. Samples of the figures used in target-searching task. The figure enclosed by red dotted line is the target figure.

questionnaire using 7-point Likert-type scales (strongly disagree = 1 to strongly agree = 7) on three emotional dimensions (tension, fatigue and annoyance) [17], which were extracted from POMS (Profile of Mood States). Finally, 3) task performance was assessed. The participants were asked to find a target figure from many similar figures correctly as soon as possible, as shown in Figure 3. Altogether, seven indicators from these three aspects were used to indicate participants' stress and fatigue status.

2.5. Procedure

Before the formal experiment, the participants were asked to do some preliminary activities to understand the experiment process. In experiment 1, the formal experiment consisted of 6 units, and each experiment unit included 5 steps (Figure 4): 1) 5-min calculation: the participants were guided by the researcher to conduct subtraction from 1895 with a step of 13. 2) Measurements of original status: after the 5-min calculation participants' physiological responses, psychological experience and task performance were measured using a hemomanometer, self-reported scales and a 1-min target-searching task, respectively. All the indicators measured in this stage were used to show participants' original status after the 5-min calculation task. 3) Restoration period: 3-min sound stimuli broadcasting for experimental groups and 3-min silence for control group were arranged after the measurement of participants' original stress and fatigue levels. Sounds and silence were completely random in six experiment units. 4) Measurements of restorative effects: the measured indicators were completely the same as in step 2. The purpose of these measurements was to reveal participants' status after 3-min sound broadcasting or silence. 5) Rest: Before the next experiment unit, there was 2-min rest, during which participants' physiological, psychological and cognitive conditions could return to the normal level. Altogether, there were six experimental units to be carried out.

Experiments 2 and 3 followed almost the same process as Experiment 1. The main difference in those three experiments was sound stimuli. In experiment 1, the sound stimuli consisted of five different sounds: flowing water sound, birdsong, footsteps, traffic noise and air-conditioner noise. In experiment 2, the sound stimuli were five temporal sequences of flowing water sound. In experiment 3, the stimuli were replaced by audio-visual stimuli, which were the combination of positive or passive sounds with good or

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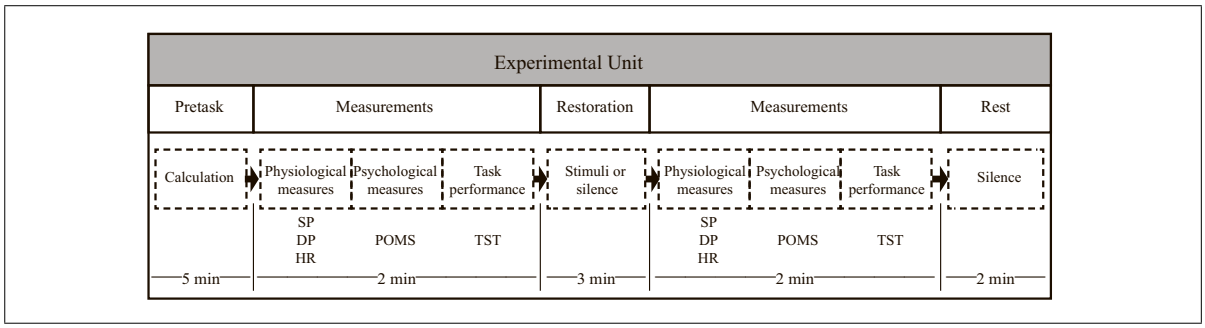

Figure 4. Diagram of experimental procedure. Each experimental unit takes approximately 15 min to be completed. The upper panel is the major steps in each experimental unit, and the details are shown in the middle panel. The bottom panel relates the measurement indicators in the procedure. SP and DP = systolic and diastolic pressure, HR= heart rate, POMS = profile of mood states, and TST = task searching test.

bad images. In the tests, binaural signals were played back by computer through headphones (AKG K702), and images were presented to participants on a large 46-inch display.

All the experiment settings were intended to simulate the situation of an open plan office.

3. Results

In all the experiments, to accommodate individual differences, each participant's raw data were transformed into difference values, and the larger difference values meant better restorative effects brought by sound stimuli or audio-visual stimuli. ANOVA analysis was conducted to detect the statistically significant mean differences among different stimuli, in terms of restorative indicators. In addition, the measurement values of each physiological response, psychological experience and task performance were compared before and after participants' exposure to sound stimuli (audio-visual stimuli in the third experiment). Paired-samples t tests were further performed to examine the difference of restorative effects before and after the restoration period.

3.1. Restorative effect of different sounds

Experiment 1 was carried out to explore whether the sounds preferred by people had a restorative effect whereas the sounds disliked by people had no such function. ANOVA analysis in Table II shows that among the seven indicators, the self-reported fatigue alleviation and annoyance attenuation were significantly influenced by various sound types, whereas in terms of other indicators, including task performance, psychological tension, and physiological responses, the effects of different sounds were almost the same. Whether the restorative effects of sounds depends on the genders of participants was examined in this study as well, and it was found that there was no significant difference between male and female participants in any indicators. In addition, Table III shows the significant correlation of subjective preference of sounds with their restorative effects on fatigue and annoyance, and compared with unpleasant sounds, the sounds preferred by people had greater effects on fatigue alleviation and annoyance attenuation.

Table II. ANOVA of sound type and gender on restorative effects. * and ** indicate significance at the 0.05 and 0.01 levels. TP, T, F, A, SP, DP, and HR represent task performance, tension, fatigue, annoyance, systolic pressure, diastolic pressure and heart rate, respectively.

	Sound type			Gender		
	F	p	η^2	F	p	η^2
TP	0.558	0.693	0.015	1.518	0.220	0.010
T	0.572	0.683	0.016	0.393	0.532	0.003
F	6.567	0.000**	0.156	0.531	0.467	0.003
A	6.074	0.000**	0.147	0.099	0.753	0.001
SP	0.387	0.818	0.010	1.462	0.229	0.010
DP	1.036	0.391	0.028	0.122	0.727	0.001
HR	0.774	0.544	0.021	0.810	0.370	0.006

Table III. Correlation between the subjective preferences (Pref) of sounds and each of the restorative indicators. * and ** mean significant correlations at the 0.05 and 0.01 levels. TP, T, F, A, SP, DP, and HR represent task performance, tension, fatigue, annoyance, systolic pressure, diastolic pressure and heart rate, respectively.

	TP	T	F	A	SP	DP	HR
Pref	-.056	.090	.366**	.352**	.072	.093	.098

Figure 5 shows the difference in participants' psychological experience before and after their exposure to various sound elements. As for the indicators of self-reported fatigue and annoyance, various sound types showed significantly different restorative effects. Air-conditioner noise, traffic noise and footsteps had almost no influence on participants' evaluation of fatigue and annoyance, which however were significantly reduced after exposure to birdsong and water sound, as shown in Table IV. Different from fatigue and annoyance, Figure 5 shows that all five sounds had restorative effects on the indicator of self-

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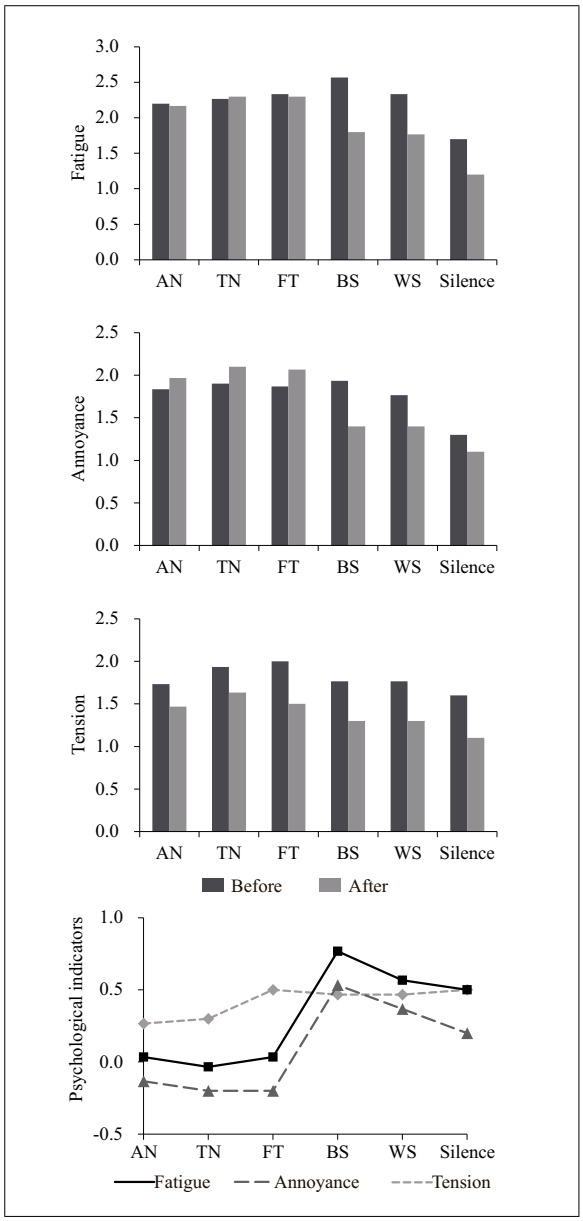

Figure 5. Effect of different sound elements on psychological experience (AN, TN, FT, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively.)

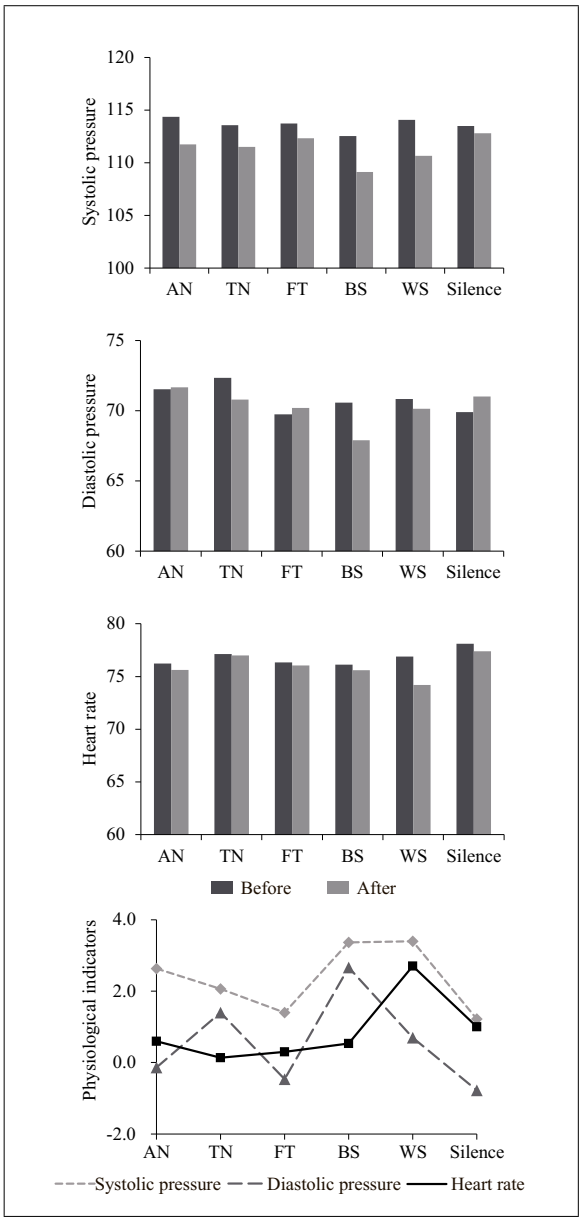

Figure 6. Effect of different sound elements on physiological indicators (AN, TN, FT, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively).

reported tension more or less. However, paired-samples t test in Table IV further indicates that only after exposure to the sounds of footsteps, birdsong, water and silence, participants were greatly relieved. Figure 5 further compares the restorative effects of sounds on fatigue, annoyance and tension, and the results show that people's fatigue was much more restored than their annoyance after their exposure to similar sounds, regardless of whether the sound was preferred by people. Moreover, birdsong showed a greater effect on both fatigue and annoyance restoration than water sound, which was better than silence.

Figure 6 shows the effects of different sound types on participants' physiological indicators, including systolic pressure, diastolic pressure and heart rate. Comparing physiological indicators before and after exposure to the sound, it seemed that all five sounds had a restorative effect on these three indicators more or less. However, birdsong and water sound were still more restorative in terms of overall physiological response compared to others, as shown in Table V. In addition, if comparing the restorative effects on systolic pressure, diastolic pressure and heart rate, it can be seen that there was no consistent tendency, and those indicators varied casually with different soundscape elements added into the environment.

In terms of task performance, it can be seen from Figure 7 that all sound types had positive effects on it and the

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Table IV. Paired-samples T test of psychological indicators before and after the sound exposure. * and ** mean significant influence at the 0.05 and 0.01 levels. AN, TN, FT, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively.

	AN	TN	FT	BS	WS	Silence
Fatigue	.813	.801	.801	.000**	.000**	.052
Annoyance	.326	.206	.083	.001**	.019*	.168
Tension	.088	.059	.000**	.001**	.006**	.015*

Table V. Paired-samples T test of physiological indicators before and after the sound exposure. * and ** mean significant influence at the 0.05 and 0.01 levels. AN, TN, FT, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively.

	AN	TN	FT	BS	WS	Silence
Systolic	.042*	.221	.188	.019*	.048*	.800
Diastolic	.924	.227	.734	.017*	.530	.460
Heart rate	.710	.912	.701	.682	.004**	.671

most restorative sound was air-conditioner noise, which was followed by silence and water sound (Table VI). However, birdsong, which was also assessed as pleasant, showed to be the worst restorative sound for working efficiency.

3.2. Restorative effect of sound sequences

Since different sound types have been proven to have different restorative potential in the first experiment, the aim of Experiment 2 was to investigate whether the temporal sequence of sound reproduction would affect its restorative effects. Flowing water sound was used as restorative sound stimuli and there were five types of sound sequences in this experiment. Excluding the heart rate, ANOVA shows no statistically significant difference among the five temporal sequences of water sound in other restorative indicators (Table VII). Although the restorative effects on heart rate showed significant difference between different sound sequences, it varied quite casually with the variation of sound sequences.

In addition, male and female participants showed significantly different restoration in terms of tension, fatigue and annoyance (Table VII). Compared with female participants, male participants attained more recovery from the original tense, annoyed and fatigued status if adding flowing water sound into the stimulated open office situation regardless of the sound sequences, as shown in Figure 8. Moreover, male participants were more sensitive to the varying of sound sequences of flowing water sound.

3.3. Restorative effect of audio-visual conditions

Many studies have proven that visual and auditive elements were both important to environmental perceptions. However, sometimes it is difficult to change both visual

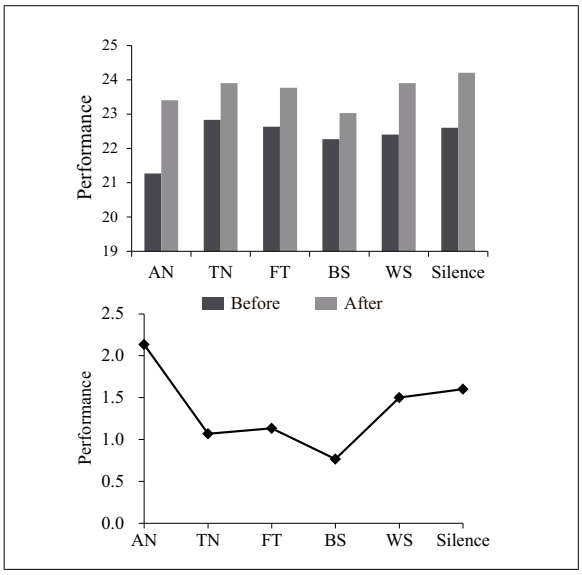

Figure 7. Effect of different sound elements on task performance (AN, TN, FT, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively).

Table VI. Paired-samples T test of task performance (Tp) before and after the sound exposure. * and ** mean significant influence at the 0.05 and 0.01 levels. AN, TN, FT, BS, and WS represent air-conditioner noise, traffic noise, footsteps, birdsong and water sound, respectively.

	AN	TN	FT	BS	WS	Silence
Tp	.005**	.162	.131	.275	.025*	.021*

Table VII. ANOVA of temporal sequence of flowing water sound and gender on restorative effects. * and ** mean significant influence at the 0.05 and 0.01 levels. TP, T, F, A, SP, DP, and HR represent task performance, tension, fatigue, annoyance, systolic pressure, diastolic pressure and heart rate, respectively.

	Sound type			Gender		
	F	p	η^2	F	p	η^2
TP	0.409	0.801	0.024	0.507	0.479	0.007
T	0.598	0.665	0.022	23.400	0.000*	0.244
F	0.189	0.943	0.008	8.986	0.004**	0.116
A	0.735	0.571	0.036	6.288	0.015*	0.083
SP	1.053	0.387	0.052	1.307	0.257	0.018
DP	1.496	0.214	0.083	0.806	0.373	0.011
HR	2.603	0.044*	0.114	0.498	0.483	0.006

and acoustic conditions in a certain environment to increase its restorative effects, particularly for open-plan offices. Therefore, the interaction of audio-visual factors on environmental restorative effects was worthy of being discussed to redesign the environment effectively. There are four audio-visual stimuli (water sound with good image, water sound with bad image, air-conditioner noise with good image and air-conditioner noise with bad image) in

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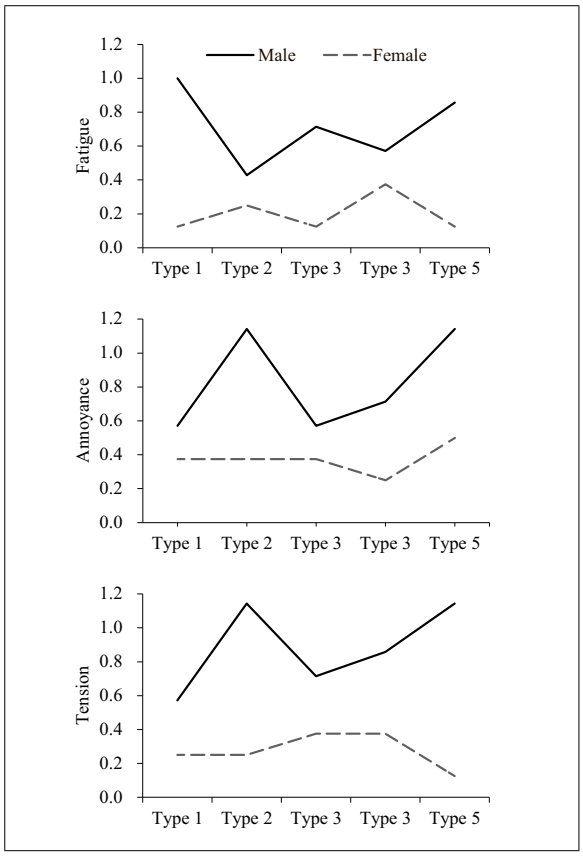

Figure 8. Effect of different sound sequences of flowing water sound on psychological experience (Type 1: continuous water sound; Type 2: intermittent water sound, 10 s of water sound and 10 s of silence; Type 3: intermittent water sound, 10 s of water sound and 5 s of silence; Type 4: intermittent water sound, 5 s of water sound and 10 s of silence; Type 5: intermittent water sound, random duration for water sound and silence).

this experiment. ANOVA analysis in Table VIII shows similar results to the first experiment, and subjective fatigue and annoyance level were significantly difference among the four audio-visual stimuli. However, this phenomenon was almost the same for both male and female participants.

The comparison of psychological experience before and after participants' exposure to different audio-visual stimuli is shown in Figure 9. It can be seen that tension was greatly reduced whenever the audio-visual conditions varied, but the restorative effects on fatigue and annoyance varied significantly with varying audio-visual conditions. Nevertheless, the combination of water sound and good images was the best restorative conditions, as all the psychological experience improved significantly, as shown in Table IX. In addition, the restoration effects of water sound were significantly better than that of air-conditioner noise regardless of the visual scene was good or bad.

As for physiological measures shown in Figure 10, there was no consistent change with various audio-visual conditions. In contrast, participants' task performance was improved after exposure to all types of audio-visual conditions, although the improvement was not significant (Table X). It can be seen from Figure 11 that participants performed better when they were exposed to consistent audio-visual conditions (AN+BI and W+GI). If audio stimuli did not match the visual stimuli well, participants had worse task performance.

Table VIII. ANOVA of audio-visual stimuli and gender on restorative effects. * and ** mean a significant influence at the 0.05 and 0.01 levels. TP, T, F, A, SP, DP, and HR represent task performance, tension, fatigue, annoyance, systolic pressure, diastolic pressure and heart rate, respectively.

	Sound type			Gender		
	F	p	η^2	F	p	η^2
TP	0.608	0.611	0.016	0.556	0.457	0.005
T	1.233	0.301	0.032	0.129	0.721	0.001
F	4.680	0.004**	0.105	0.344	0.558	0.003
A	9.416	0.000**	0.195	0.746	0.390	0.005
SP	1.021	0.386	0.026	0.142	0.707	0.001
DP	2.002	0.118	0.050	0.116	0.734	0.001
HR	1.332	0.268	0.034	0.074	0.785	0.001

Table IX. Paired-samples T test of psychological indicators before and after the audio-visual exposure. * and ** mean significant influence at the 0.05 and 0.01 levels. AN and W represent air-conditioner noise and water sound, respectively, BI and GI represent bad image and good image, respectively.

	AN+BI	W+BI	AN+GI	W+GI
Fatigue	.625	.001**	1.000	.016*
Annoyance	.184	.054	.476	.000**
Tension	.017*	.000**	.005**	.000**

Table X. Paired-samples T test of physiological indicators and task performance before and after the audio-visual exposure. * and ** mean significant influences at the 0.05 and 0.01 levels. AN and W represent air-conditioner noise and water sound, respectively, BI and GI represent a bad image and good image, respectively.

	AN+BI	W+BI	AN+GI	W+GI
Systolic	.012*	.007**	.095	.281
Diastolic	.088	.024*	.781	.903
Heart rate	.931	.828	.037*	.736
Task perf.	.682	.491	.467	.066

4. Discussion

The aim of this study is to explore the restorative effects of soundscape elements on physiological responses, psychological experience and task performance in a simulated open-plan office. As for the first experiment which was performed to investigate the restorative effects of different sound types, results shows that participants' annoyance and fatigue restored more significantly after expo-

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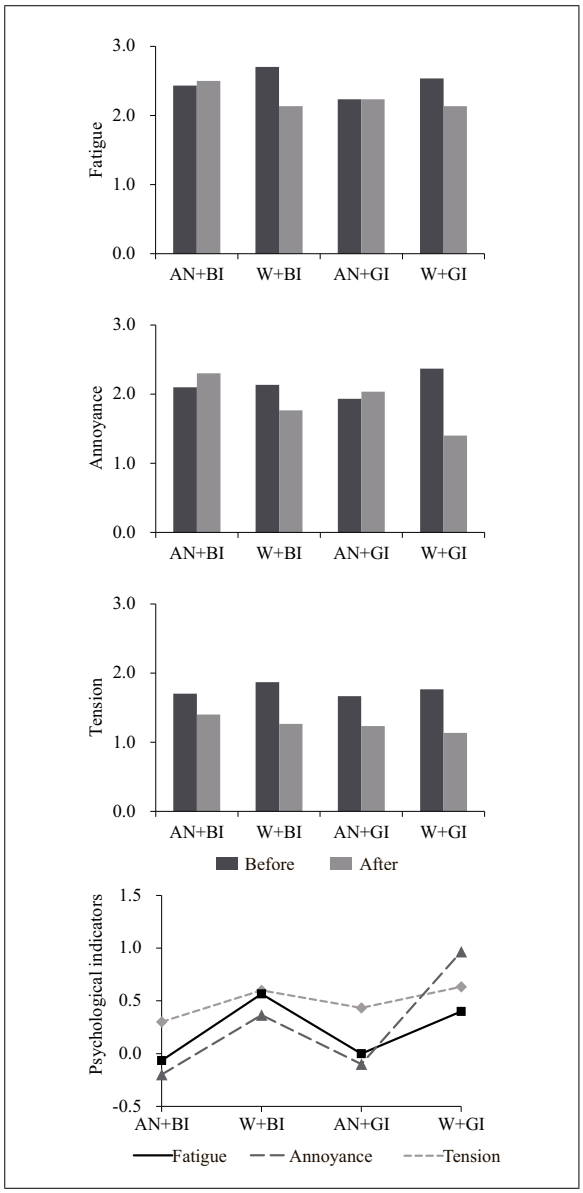

Figure 9. Effect of different audio-visual conditions on psychological experience (AN and W represent air-conditioner noise and water sound, respectively; BI and GI represent a bad image and good image, respectively).

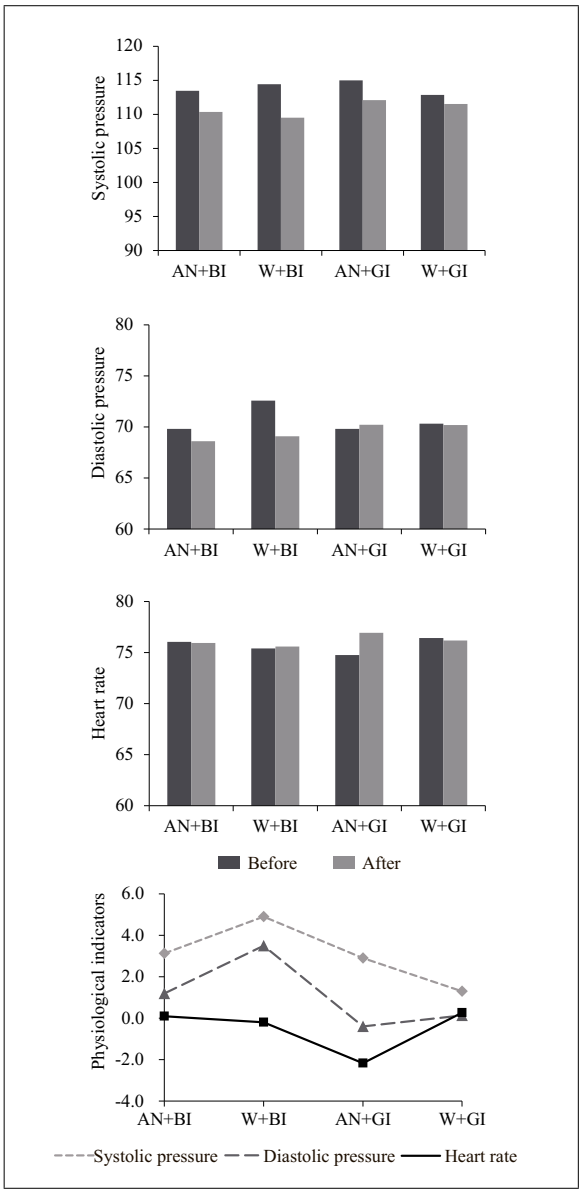

Figure 10. Effect of different audio-visual conditions on physiological indicators (AN and W represent air-conditioner noise and water sound, respectively; BI and GI represent a bad image and good image, respectively).

sure to the flowing water sound and birdsong rather than the other three sounds, which were footsteps, traffic noise and air-conditioner noise. Because water sound and birdsong were preferred by people and considered as good soundscape elements among these five sounds, it could be concluded that adding good soundscape elements into the environment could improve the restorative effects of the environment at least for the restoration of annoyance and fatigue. This result supports previous studies indicating that restorative environments are highly associated to subjective preference [29]. In addition, statistical analysis showed that the restorative effects of soundscape elements were mainly shown in the psychological experience rather than physiological responses. This was different from the hypothesis that exposure to pleasant soundscapes would lead to a better restoration on physiological measures compared to unpleasant sounds according to previous studies [20, 24]. This is perhaps because physiological response was quite instantaneous and easily influenced by intrinsic and extrinsic factors. Moreover, air-conditioner noise which was perceived as an unpleasant sound was shown to be much more helpful for participants' task performance compared to other pleasant sounds. A possible reason was that people who were used to work in an open-plan office had become used to the air-conditioner noise, but the improvement in task performance was accompanied by fa-

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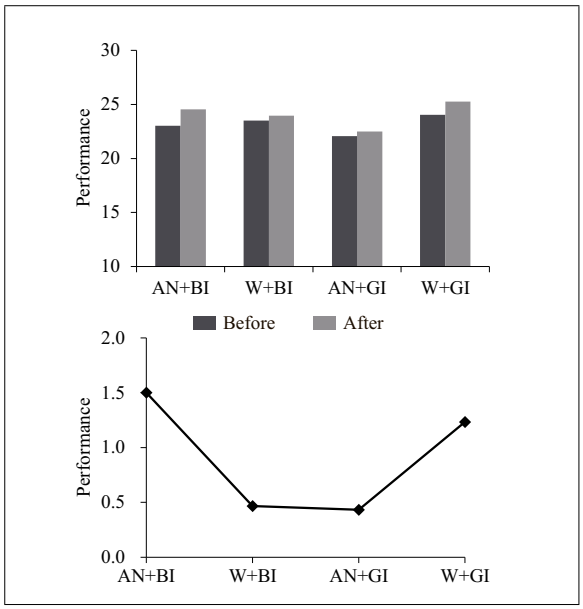

Figure 11. Effect of different audio-visual conditions on task performance (AN and W represent air-conditioner noise and water sound, respectively; BI and GI represent a bad image and good image, respectively).

tigue and annoyance. In addition, the continuous and uniform air-conditioner noise could rarely cause mood swings and distractions. In contrast, birdsong, which was assessed as a pleasant sound, showed to be the worst restorative sound for working efficiency. It was reasonable as birdsongs have larger fluctuation strength, which can easily divert people's attention from work [30].

During the second experiment, flowing water sound was used to explore the restorative effects of different sound sequences. There was no significant difference was found among the five temporal sequences of water sound in terms of restorative indicators except heart rate. However, what was the specific factor that affected people's heart rate still needs further study. All in all, whether the added sound was constant did not influence its restorative effect, and adding a good soundscape element discontinuously might be an efficient method for environmental improvement. Additional, the results shows male participants got better restoration than female participants after exposure to different sound sequences in terms of psychological experience. It means that deliberate soundscape design was necessary for the open-plan offices where the majority of employees were male.

Finally, the third experiment investigated the restorative effects of different audio-visual conditions. It was interesting that the restoration effect of flowing water sound was significantly better than that of air-conditioner noise regardless of the visual scene was good or bad. This phenomenon was very apparent for the restoration of psychological experience. In other words, sound conditions might have a greater effect on the restoration of people's fatigue and annoyance, at least in the simulated open-plan offices. Therefore, adding a suitable soundscape element was a better choice than changing the visual scene for employees' recovery of fatigue and stress. This result was consistent with Preis's study that suggested that audio input in environmental assessments had a stronger influence than visual input [26]. Moreover, the results shows that congruent audio-visual conditions had better restorative effects than those conditions where audio stimuli and visual stimuli did not match. It was perhaps because people felt more conventional when the sound and the image were both positive or both negative. This is also consistent with previous studies indicating the importance of the congruence of acoustic stimuli and visual stimuli on environmental perception [31].

5. Conclusion

In the simulated situation of an open-plan office, three experiments were carried out to explore the restorative effects of soundscape elements. Based on the physiological response, psychological evaluation and cognitive performance, the following conclusions could be drawn.

1. Compared with sounds perceived as unpleasant, a good soundscape element perceived as pleasant, such as birdsongs and flowing water sound, had good restorative effects on people's psychological evaluation, particularly for self-reported fatigue and annoyance mitigation. In addition, a good soundscape element preferred by people had more apparent restorative effects on psychological experiemce rather than physiological response and cognitive performance.
2. As for the same soundscape element, continuous sound did not show a better restorative effect than intermittent sound. Therefore, the sound sequence in open-plan offices could be decided by other considerations, such as efficiency.
3. The restorative effects of combined audio-visual conditions on fatigue and annoyance were also significantly different between different stimuli. The most important thing was that compared with visual scenes, sound factors had greater effects on people's recovery from fatigue and annoyance in certain situations, though more systematic research was still needed.

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